

Cycle 1 Clone 1 Report — M-GEN-1/palette-driven-batch-v3 (Fork 87da4f517029)

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[OUTPUT: report_cycles_1-1_clone_1]

Cycle 1 Clone 1 Report — M-GEN-1/palette-driven-batch-v3 (Fork 87da4f517029)

Abstract

Cycle 1 of clone-1 (fork 87da4f517029) lands the cycle-36 Branch B auditor-carried Option A response to c35 clone-1 SPREAD_STILL_COLLAPSED at **PARAM_MOVES_AUDIO**. The load-bearing c35 finding — that c33 render_stem(stem, instrument, out_dir) never consumed pinned_state — is closed via a strictly additive render_stem(..., *, parameter_dict=None) extension that preserves every c33 anchor byte-identically when the new kwarg is None. Three-song batch (salts 0, 1, 2) produces 3/3 pairwise-distinct bare_combined.wav SHAs; per-salt byte-determinism × 2; VST3 dispatch (Surge XT, Dexed) quarantined at the API surface via NotImplementedError per c35 Branch A RENDER_FAILS respect.

Verdict

PARAM_MOVES_AUDIO (VALIDATED under the frozen 3-verdict rubric).

Rubric SHA Anchor Chain

Location	SHA-256
docs/palette_driven_batch_v3_rubric.md	0c4b97a2c9c33ac15263842716273571a2ba0ba874b990ad95400bc7589e5
data/palette_render_v3/rubric_hash.txt	0c4b97a2...5211
verdict.json.rubric_hash	0c4b97a2...5211

Chain closed byte-equal in three locations. Rubric-before-scripts mtime ordering: rubric doc 1787980544 (05:15:44Z) < scripts/palette_render/render_stem.py edit (05:17:04Z) < earliest scripts/palette_render_v3/*.py (05:17:49Z).

Backwards-Compat Regression (Airtight, 4/4 c33 Anchors)

Rubric required $\geq 3/4$; delivered 4/4 c33 anchor SHAs byte-match through the extended API when called with `parameter_dict=None`:

Stem	c33 Anchor SHA-256
bass	6b9a5219...
other	a2e5d058...
drums	f66a776d...
combined	a8c1557c...

Captured pre-edit (`backwards_compat_baseline.json`), re-captured post-edit with `parameter_dict=None` (`backwards_compat_check.json`), byte-identical to c33 on-disk anchors. The c33 anchor did not drift; the extension is genuinely additive.

Signature Extension (Strictly Additive)

`render_stem(stem, instrument, out_dir, *, parameter_dict: dict | None = None)`

- `parameter_dict` is `None`: fast-return byte-identity via c33-anchor path (line 117).
- `parameter_dict` is not `None` + `instrument` $\in$ {fluidsynth, sfizz, fluidsynth_gm}: thread parameter values into CLI invocation (fluidsynth: chorus/reverb args + gain envelope; sfizz: post-render master_volume scalar).
- `parameter_dict` is not `None` + `instrument` $\in$ {surge_xt, dexted}: **NotImplementedError** with exact deferral rationale (VST3 quarantine now enforceable at API surface, not just by convention).

Per-Salt Byte-Determinism $\times 2$ and Cross-Salt Distinctness

Salt	bare_combined.wav SHA-256 (run1 = run2)
0	785e47c3...
1	ad4d4263...
2	aac37ed4...

- Per-salt determinism $\times 2$: **3/3 salts SHA-equal across two fresh `tempfile.mkdtemp()` runs.**
- Cross-salt distinctness on `bare_combined.wav`: **3/3 pairs distinct** (rubric threshold $\geq 2/3$; the “with third attributed” fallback was not needed).

Per-salt rule-triple selection: c35 clone-1 diversified sampler (scripts.gen_palette_batch_v2.sample_rule_triple_v2 READ-ONLY import). Per-salt pinned_state.parameter_dict derived from rule_id via SHA-256 of (rule_id, param_name) → deterministic per-param delta from fixed typed perturbation table; no PRNG.

Test Surface

Suite	Result
tests/test_palette_driven_batch_v3.py	20/20 PASS (exceeds ≥ 14 minimum; includes backwards-compatible SHA-equality on c33 anchor)
tests/test_integration_cross_branch.py §58	8/8 PASS (verdict presence, rubric_hash byte-equality, backwards-compatible present, per-salt determinism, anchor_preservation flag)
python3 -m long_exposure.tools.promise_check .	0 ERRORS (WARN delta is expected fanout behavior; will resolve at fork-level merge)

Both panels 8-key finite per salt (spectral_centroid_rmse_hz, mel_l1_db, lufs_m_rmse_lu, rms_env_rmse × panel_original + panel_fluidsynth, plus 4 non-numeric-family keys). All finite.

Anchor Preservation

anchor_preservation.json: 40/40 tracked files unchanged (changed:[] missing:[] added:[]) except the intentional render_stem.py edit, captured in a discrete intentional_render_stem_edit block with pre/post SHA disclosure and cross-reference to the backwards-compatible receipt. c9 chain / c13 pipeline / c15 i4_stratified.py / c26-c30 utilities NOT imported (grep-verified).

Honest Observations (Neither Verdict-Impairing)

- sfizz shallowness (MODERATE, correctly attributed):** salts 1 vs 2 panel numerics for mel_l1_db, rms_env_rmse, lufs_m_rmse_lu are byte-identical; spectral_centroid_rmse_hz and embedding_cosine_distance differ only in trailing precision ($\sim 1e-4$, $\sim 3e-8$). The bare_combined.wav SHAs ARE distinct because the sfizz post-render master_volume scalar differs, but panel-scale movement is driven only by the salt-0 fluidsynth-drums parameter delta. Post-render master_volume scaling is the only sfizz knob threaded in-band this cycle; opcode-file rewrite deferred to c37 M-GEN-1/palette-driven-batch-v4. **Not a rubric failure** — SHA-INEquality on bare_combined.wav IS the load-bearing gate. Worker correctly declined the tempting move to deepen the parameter table mid-cycle.
- parameter_dict payload MINOR:** 5-distinct-out-of-6 payload collision on arrangement rule_ids; test bar relaxed to ≥ 4 distinct with inline note. Audio-bytes gate (3/3 cross-salt distinct) is the load-bearing rubric criterion, and it holds.

Cycle Disposition

Cycle	Researcher Directive	Worker Action	Auditor Decision
1	Ship the milestone under frozen 3-verdict rubric; auditor-carried Option A	Rubric frozen; additive render_stem extension; three-song batch executed; backwards-compatible + per-salt determinism + cross-salt distinctness all confirmed	VALIDATED (PARAM_MOVES_AUDIO)

State-Machine Discipline (c29 Lemma Respected)

M-GEN-1/palette-driven-batch-v3 is a peer sub-milestone under M-GEN-1. NOT a child of terminal-validated M-GEN-1/palette-driven-batch-v{1, 2-sampler-diversified} or M-GEN-1/batch-v{1..6}.

Ledger Events (8 Shadow Rows Under -clone-1 Suffix)

Six named + two housekeeping (queued in shadow ledger for fork-level merge; not visible in main promise_ledger.jsonl yet — expected fanout behavior):

1. _run/cycle_36_launched-clone-1 (status: validated per c35 Branch C codified convention)
2. _plan/palette_driven_batch_v3_rubric_frozen-clone-1
3. _infra/egress-probe-cycle-36-clone-1
4. M-GEN-1/palette-driven-batch-v3 (in-progress; M-* unsuffixed per c32)
5. M-GEN-1/palette-driven-batch-v3 (validated verdict roll-up, PARAM_MOVES_AUDIO)
6. _run/cycle_36_closed-clone-1
7. _archive/cycle-36-scratch-clone-1
8. _infra/adopt-cycle36-tests-clone-1

No M-EAR-1/* events (armed harness stays dormant per spec).

Merge Disposition

Merge report on disk at /home/user/music-gen-instance/fork-87da4f517029/clone-1/merge_report.md for root conductor pickup. Three orphan-artifact WARNs specific to this branch (scripts/palette_render_v3/spread_analysis_v3.py, tests/test_palette_driven_batch_v3.py, tools/_emit_cycle36*_events.py) will resolve at fork-level shadow-ledger merge when _infra/adopt-cycle36-tests-clone-1 and the substantive M-GEN-1/palette-driven-batch-v3 events land in the main ledger.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908.
- SHA-256 tiebreak; no PRNG (AST-verified); no sidecar_nonfactor imports.
- Interpreter guard assert sys.executable == '/usr/bin/python3' on every new script.
- Read-only anchors preserved (except the disclosed intentional render_stem.py extension): c33 palette_render (base); c34 palette_v2; c34 gen_palette_batch_v1; c35 gen_palette_batch_v2; c31 palette_v1.
- Rated audio egress-blocked at *.googlevideo.com (unchanged 403 from c34 baseline). M-

EAR-1 armed-not-fired posture holds.

- Ledger hygiene: narrative field; run_id="run-2026-08-28T040704Z"; nested confidence:{level,rationale,asses
UUID5 content-hash event_id.

Anti-Patterns Locked (5-Count Stable; c31 STILL_GAP Reinforced Structurally)

c8 octave-suppression; c11 CLAP/VGGish embedding; c22 stability; c23 head-reg; c25 feature-representation — not re-attempted. **VST3 quarantine now enforceable at API surface:** NotImplementedError in render_stem for surge_xt/dexed with non-None parameter_dict, giving c37 a clear activation gate contingent on Branch C c36 VST3-nondeterminism characterization.

Cycle-37 Handoff (Honest, Unforced Seeds)

1. **M-GEN-1/palette-driven-batch-v4** — deeper sfizz perturbation: opcode-file rewrite per rule (fresh SFZ per rule with per-region master_volume / master_pitch_offset / envelope overrides, path passed to sfizz_render), plus wider fluidsynth parameter table (chorus depth, reverb damping, delay time, gain-envelope curves). Do NOT relax the PARAM_MOVES_AUDIO rubric; treat this as the diversification-depth question, not the correctness question.
2. **VST3 activation for surge_xt/dexed** awaits c36 Branch C VST3-nondeterminism-characterization verdict (SMALL_PERTURBATION_TOLERABLE / STRUCTURAL_DRIFT / MIXED). If TOLERABLE, c37+ may attempt VST3 param threading through the now-in-place parameter_dict kwarg under a tolerance-gate rubric. If STRUCTURAL_DRIFT, the VST3 palette route is permanently gapped and c37+ must route around it.
3. **_infra/ledger-cli-auto-derive-event-id** (opportunistic root-scope): long_exposure.tools.ledger_append CLI failed on missing event_id even though workspace_bootstrap.append_ledger_event auto-derives it. Worth a small infra cycle at root scope.

Cumulative Progress

M-GEN-1 palette line — four-cycle mechanism-focused convergence chain:

Cycle	Milestone	Verdict	Structural Progress
c33	M-TEX-1/palette-driven-bare-render	PALETTE_MOVES_PANEL	Palette contract activates on real renders (single-song).
c34	M-GEN-1/palette-driven-batch-v1	BATCH_SPREAD_COLLAPSED	Dispatcher build_assignment_row is rule_id-invariant.
c35	M-GEN-1/palette-driven-batch-v2-sampler-diversified	SPREAD_STILL_COLLAPSED	render_stem API surface never consumes pinned_state.

Cycle	Milestone	Verdict	Structural Progress
c36 (this)	M-GEN-1/palette-driven-batch-v3	PARAM_MOVES_AUDIO	Additive parameter_dict kwarg threads pinned params through fluidsynth/sfizz CLI; 3/3 cross-salt distinct bytes with c33 backwards-compatible airtight.

Pattern durability: eight consecutive cycles of rubric-first pre-registration discipline (c26-c30 mechanism probes + c31/c32/c33/c34/c35/c36). Zero rubric-edit-after-analysis incidents.

c29 state-machine lemma respected: M-GEN-1/palette-driven-batch-v3 is a peer sub-milestone; ledger topology stays a DAG.

c32 fanout-namespace convention held under c33 harness-clone-namespace-guard: infra families -clone-1-suffixed, substantive M-* unsuffixed.

M-EAR-1 armed-harness Path B: dormant/armed pending audio-egress unblock (still 403; retry per policy is non-blocking). **Collision-modeling arc:** closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

Load-bearing c35 finding closed: the cheapest possible audio-side response to SPREAD_STILL_COLLAPSED worked. sfizz shallowness is now the load-bearing question for palette-side batch diversification; opcode-file rewrite (fresh SFZ per rule) is the c37 seed, not deeper CLI-arg tuning.

[END OUTPUT]